Effects of State Dependent Robot Behavior Adaptation on Trust and Team Performance in Human-Robot Teams

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In this paper, we study the effects of robot behavior adaptation through a state-dependent reward function in a trust-aware Markov Decision Process framework where the robot acts as an action recommender to the human in a human-robot collaborative Intelligence, Surveillance, and Reconnaissance (ISR) Mission. Toward this, we first present a framework for learning a state-dependent reward function through human data by querying them about action choices in the absence of recommendations. We use this framework and collect extensive data through Amazon Mechanical Turk to learn this state-dependent reward function. In a follow-up study, we then compare a robot interaction strategy using this state-dependent reward function with a baseline robot strategy using a constant reward function in two contexts: A low vulnerability context where the human-robot team has a lot of resources to expend during the mission and a high vulnerability context where the human-robot team has a much more limited pool of resources. We compare these strategies in terms of various trust, reliance, and team performance measures. Our results indicate that the state-dependent strategy is able to garner more trust from the humans, is rated higher in terms of future reliance intentions, and performs better objectively compared to the constant strategy. These effects are further enhanced in the high vulnerability case. These results show the effectiveness of using fine-grained reward functions by the robot during interactions with humans in terms of trust and team performance, especially when the vulnerability to the human is high.

CCS Concepts: • **Human-centered computing**; • **Computer systems organization → Robotic autonomy**; • **Information systems → Decision support systems**;

Additional Key Words and Phrases: Trust in HRI, Human-Robot Teaming, Trust-aware Decision-Making, Adaptivity

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1 Introduction

To facilitate smooth and effective human-robot interaction, trust is a very important factor [21, 34, 35, 46] The area of research concerning trust-driven human-robot interaction has seen an enormous rise in popularity in recent years [8, 25, 37, 62]. Further, responsivity in autonomous agent behavior

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Chiou and Lee [17] has been postulated to be almost as important, if not more, than reliability in the context of human-autonomy teaming. In this work, we try to frame responsivity of autonomous systems through a state dependent reward function perspective and compare it with another system using a constant (non state dependent) reward function. We study these differences in the context of a "robot recommender - human decision-maker" scenario wherein the robot tries to maximize the reward function over the planning horizon. Our results indicate that a robot using such a state dependent reward function results in higher trust and better team performance compared to the robot using a constant reward function, especially in a context where the vulnerability of the human is high.

Our contributions in this paper include:

- We present a framework for learning a state dependent reward function in a dual objective human-robot collaboration task where the robot acts as an action recommender to the human. We demonstrate this framework through a human-subjects study done over Amazon Mechanical Turk. The learned reward function shows the expected behavior of risky and risk-averse behavior in different contexts.
- We compare two interaction strategies - one using the learned state dependent reward function and the other using a constant reward function - on trust and team performance through an in-person human-subjects study in two separate contexts - a low vulnerability context and a high vulnerability context. Our results indicate that the state dependent robot strategy leads to higher trust and better team performance in general. These effects are further boosted in the high vulnerability context.

The rest of the paper is organized as follows: Section 2 discusses prior literature that motivates our work. Section 3 presents details of our mathematical model of human-robot interaction in a context where the robot acts as an action recommender to the human. Section 4 details the first study within this work, dealing with learning a data-driven state dependent reward function through data collected using Amazon Mechanical Turk. Section 5 presents details about the second study where we compare a robot using this state dependent reward function with a robot using a constant reward function through an in-person human subjects experiment. Finally, Section 6 discusses the implications of our results and some limitations and directions of future work.

2 Related Work

There are three major areas of research related to our work. We discuss seminal works and recent advances in these fields below.

2.1 Human-Autonomy Teaming

Human-autonomy teaming (HAT) [43, 49, 61] has attracted significant attention across robotics, artificial intelligence, and human factors domains, motivated by the potential of combining human flexibility and judgment with the efficiency and scalability of autonomous systems. Studies have explored various aspects of HAT, including trust calibration [3], decision support [8], communication protocols [32], and shared situation awareness [13].

Ma et al. [44] argue that in scenarios where multiple humans and robots team up to perform a common task, traditional HRI metrics of performance and efficiency should be expanded to include teamwork metrics. Towards this, they propose five distinct metrics that capture ecological and cognitive aspects of teamwork.

Liu et al. [40] propose a resource coordination algorithm that enables robots to implement robust schedules that actively characterize the performance of their human teammates. Their user studies indicate that human performance is maximized when the robot dedicates more time to exploring the capabilities of the human teammates.

Robinson et al. [56] tested human-robot teams vs robot only teams in 16 real world missions based on the DARPA Subterranean Challenge. Their results indicated that human-robot teams performed significantly better in the presence of errors or unforeseen circumstances that lead to the robot failing without intervention from a human operator. Pérez-D'Arpino et al. [53] experimentally verified the effectiveness of a human-robot teaming approach versus a direct control approach for teleoperating a robotic arm. Zhao et al. [69] argue for the need for adaptivity of autonomy to individuals within a group and discuss the challenges for collaborative autonomous systems that will interact with various individuals. O'Neill et al. [52] present an excellent review of empirical research conducted in the area of HAT.

These studies provide motivation toward making human-robot teams more effective and in this work, we try to do just that by using a data-driven state-dependent reward function for a recommender robot which allows for real-time context-dependent adaptation of recommendations leading to better trust and team performance. These results are in agreement with prior literature [17], where the authors promote the notion of responsivity of autonomous systems which influences patterns of engagement and cooperation, through trust development.

2.2 Reward Learning in HRI

Reward learning in human-robot interaction (HRI) seeks to bridge the gap between human intent and robot behavior by enabling robots to infer [50], adapt to [9], and optimize for human preferences [12, 51], demonstrations Mehta and Losey [47], corrections Losey et al. [41], and goals [39]. This area has gained substantial momentum in recent years as researchers strive to make autonomous systems more intuitive, safe, and effective in collaborative settings [6].

The field started with the seminal paper by Ng et al. [50] which introduced the concept of inverse reinforcement learning: learning reward functions for autonomous agents through demonstrations by human experts. After that, many different algorithms have been presented to solve the inverse reinforcement learning problem [2, 55, 70].

Hadfield-Menell et al. [30] introduced the concept of Cooperative Inverse Reinforcement Learning (CIRL) wherein a human and a robot are working as a team to maximize the human's reward function. The robot initially does not know the reward function but tries to learn it through interactions with the human. Fisac et al. [22] propose a solution to the CIRL problem under the assumption that the human acts pedagogically in order to teach the robot the underlying reward function.

Another area of research that has been gaining momentum in recent years is that of active learning of reward weights [7, 11, 57]. The objective here is for the robot to ask "smart" queries to the human in order to gain the most information about the human's underlying reward function through their answers to the queries. We utilize a part of this concept in order to emphasize the difference between the two interaction strategies we use for our recommender robot.

In our previous works, [9, 10] we utilized Bayesian Inverse Reinforcement Learning [55] to infer human rewards during interaction. Our results indicated that in the presence of an informed prior on the reward weights, further personalization does not improve team performance or trust in the robot. However, we had a couple of limitations in this study. First, we assumed that the reward function is constant throughout the interaction. However, this may not be the case, as the human's preferences, and hence their reward functions, may change over time and depending on the context. Second, we only considered a fixed starting state for the human-robot team, leading to a limited exploration of the state space. We theorize that having a state-dependent reward function for the robot will lead to higher trust and better team performance when more of the state space is explored. We try to address these limitations through our current work.

2.3 Trust Driven HRI

Human trust on the robot [36] has emerged as a critical factor in HRI [21]. Trust influences the willingness of humans to rely on [1, 34, 59], collaborate with [14, 24], and accept recommendations [4, 9] from robotic systems.

In order to facilitate smooth and effective collaboration between humans and robots, the robots must be equipped with the ability to estimate the human's trust during the interaction, which may change over time Chi and Malle [16]. Toward this, a plethora of dynamic trust models have been developed over the years. Xu and Dudek [65] proposed a dynamic-Bayesian network based quantitative trust model for a "human supervisor-robot worker relationship". Guo and Yang [26] proposed a beta-distribution based dynamic trust model that can be personalized for each person that interacts with the robot through a training set. The parameters can then be occasionally updated based on feedback from the person. Guo et al. [28] extend this work for a multi-human multi-robot scenario. Dagdanov et al. [20] improve the beta distribution model by considering fine-grained timescales and a maximum-entropy based reward specification to constitute the required performance metric. Williams et al. [64] use a Partially Observable Markov Decision Process (POMDP) without a reward function to model the coupled dynamics of human trust and self-confidence. The model is calibrated through human data, and showcases relationships between self-confidence, trust, and reliance on automation.

Having an estimate of trust is one half of the problem. The other half of the problem is to enable trust-aware behavior adaptation for the robot. A substantive amount of research has also been done toward the solution of this problem in recent years [4, 27, 37, 38, 62, 68]. Xu and Dudek [66] propose the Trust-Aware Conservative Control (TACtiC) which maintains a belief over the trust state of the human operator and changes robot behavior to be more conservative whenever a loss in trust is detected. Chen et al. [14, 15] develop a trust-POMDP model to incorporate trust into robot decision-making. The solution to their trust-POMDP results in robot behavior that improves team performance by effectively modulating human trust.

Mangalindan et al. [45] study the robot-worker, human-supervisor scenario in which the robot is tasked with picking up objects in the environment. They propose an interaction strategy for the robot that dynamically decides whether to seek help from the human supervisor based on their level of trust on the robot. Their results show that this proposed policy performs better than a trust-agnostic baseline policy.

Alzahrani et al. [5] take a reinforcement learning approach towards optimizing human trust when interacting with robots in simulated environments. Their approach resulted in an interaction model that dynamically adjusted trust based on task outcomes to enhance performance and reduce the risks of extreme over- and under-trust.

Guo et al. [25] and Bhat et al. [8] use a trust-aware MDP framework using the beta distribution based dynamic trust model [26] along with a human trust-behavior model to generate optimal recommendations from a robot. We extend this work by considering a more fine-grained reward function that incorporates the current state of the human. We compare a robot using this reward function to a robot using a constant reward function that was earlier shown to lead to high trust and team performance [9].

3 Mathematical Modeling

We examine a scenario in which a human sequentially inspects N sites in a town to search for potential threats. The human has two action choices: either breach the site directly or use a robotic armored rescue vehicle (RARV) for protection. Prior to each decision, an intelligent agent provides action recommendations to the human.

Directly breaching a site is quicker but involves risk—the human may be harmed if a threat is present. In contrast, deploying the RARV takes additional time but guarantees safety from harm should a threat be encountered. Harm and time expenditure are quantified as point deductions from the team's score. The human begins with a health level H and a time level C. If the human faces a threat without the RARV, they lose h health points. Each use of the RARV results in a loss of c time points.

We model the interaction between the human and the intelligent recommender agent as a fixed-horizon trust-aware Markov Decision Process (MDP) [8] consisting of states, actions, rewards, transition function, and a human behavior model.

3.1 Trust-Aware Markov Decision Process

3.1.1 States. The trust of the human on the intelligent agent t, the remaining health points H, and the remaining time points C.

3.1.2 Actions. The two actions available to the human: USE (1) or NOT USE (0) the RARV.

3.1.3 Reward function. Since the objective is to minimize mission time *and* minimize damage to the human, the reward function is modeled as a negative convex combination of the cost for losing health and the cost for losing time. Specifically,

$$R(H, C, a, D) = -w(H, C)\mathcal{H}(a, D) - (1 - w(H, C))\mathcal{C}(a). \tag{1}$$

Here, $w(H, C)$ is the reward weight for losing health, $\mathcal{H}(a, D)$ is the cost for losing health, $\mathcal{C}(a)$ is the cost for losing time, a is the action selected by the human, and D is a binary random variable indicating the presence of threat.

$$\mathcal{H}(a, D) = hD(1 - a), \tag{2}$$

$$\mathcal{C}(a) = ca, \tag{3}$$

where h is the cost for losing health and c is the cost for losing time. For our study, we set $h = c = 10$.

3.1.4 Transition function. We use the beta distribution trust model [26] with the reward-based performance metric [8] as the transition function for trust. Specifically, trust at the i^{th} interaction, t_i is modeled as a beta distributed random variable with parameters α_i and β_i which are updated based on positive and negative experiences with the recommendations.

$$t_i \sim Beta(\alpha_i, \beta_i) \text{ where} \tag{4}$$

$$\alpha_i = \alpha_{i-1} + p_i v^s, \tag{5}$$

$$\beta_i = \beta_{i-1} + (1 - p_i)v^f. \tag{6}$$

We note that $(\alpha_0, \beta_0, v^s, v^f)$ are the personalized trust parameters associated with every individual that interacts with the intelligent agent, p_i is the reward-based binary performance of the recommendation at the i^{th} search site. This performance metric is mathematically represented as:

$$p_i = \begin{cases} 1, & \text{if } R(H, C, a_i^r, D) \geq R(H, C, 1 - a_i^r, D), \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

Here, R is the reward function (Eq. 1), a_i^r is the recommended action at the i^{th} search site.

We update the health and time according to the rules mentioned above. The human loses health when encountering a threat without protection from the RARV. The human loses time for using the RARV.

$$H_{i+1} = \begin{cases} H_i - h, & \text{if } a_i^h = 0 \text{ and } D_i = 1, \\ H_i, & \text{otherwise.} \end{cases} \tag{8}$$

$$C_{i+1} = \begin{cases} C_i - c, & \text{if } a_i^h = 0, \\ C_i, & \text{otherwise.} \end{cases} \tag{9}$$

Here, i is the site index, H_i and C_i are the health and time points remaining before the i^{th} interaction, a_i^h is the action chosen by the human at the i^{th} site, and D_i is a binary variable indicating the presence of threat at the i^{th} site.

3.1.5 Human behavior model. We use the Bounded Rationality Disuse Model of human behavior when interacting with recommender systems [9]. It states that the human chooses the recommended action with the probability equal to their dynamic trust on the recommendation system. With the remaining probability, the human chooses an action using the bounded rationality model, which means choosing an action with a probability proportional to the exponential of the expected reward associated with that action. Overall, for our case, the probabilities are:

$$P(a_i^h = a | a_i^r = a) = t_i + (1 - t_i) p_i^a, \tag{10}$$

$$P(a_i^h = 1 - a | a_i^r = a) = (1 - t_i)(1 - p_i^a). \tag{11}$$

Here, a_i^h is the action selected by the human, a_i^r is the action recommended by the robot, and t_i is the human's level of trust on the recommender at the i^{th} search site and p_i^a is defined as:

$$p_i^a \propto \exp\left(\kappa E[R(H_i, C_i, a, D_i)]\right), \tag{12}$$

where κ is the "rationality coefficient" of the human. Its value controls the level of randomness in the human's action choices: a lower value implies that the human is making more random action choices while a higher value implies that the human is picking the optimal action choice more often (behaving rationally). The robot solves the MDP using finite-horizon value iteration to generate its optimal recommendation.

4 Study 1 - Learning the Reward Function

This section describes the first phase of our overall study - data collection for learning the state dependence of reward weights. We used Amazon Mechanical Turk (MTurk) for collecting data. Our main insight was that assuming the bounded-rationality model of human behavior and the absence of recommendations, we can find the tipping point of the threat level at which a majority of the population will change their choice of action from risk-taking to risk-averse. This threat level (d^*) corresponds to the threat level at which the one-step expected reward for the human for both the actions is the same. Mathematically,

$$\begin{aligned} & E_D\left[R(H, C, 0, D)\right] = E_D\left[R(H, C, 1, D)\right] \\ \implies & w(H, C) h E[D] = (1 - w(H, C)) c \\ \implies & d^* = \frac{(1 - w(H, C)) c}{w(H, C) h}. \end{aligned} \tag{13}$$

So, the idea is to collect action-choice data for a set of states of (H, C) for a series of threat levels. We can then train a logistic regression model to find d^* and thus $w(H, C)$ from Eq. 1. Fig. 1 shows an example of such a query.

Fig. 1. A sample query asked to an MTurk worker. The worker was shown this information and asked whether s/he would choose to use the RARV or not.

4.1 Data Collection

We defined 6 health states, 6 time states, and 11 threat levels, resulting in a total of 396 distinct queries for MTurk workers. These queries were randomly assigned into 12 separate surveys, each containing 33 queries, to ensure the surveys could be completed within a reasonable time. At the start of the survey process, participants were asked to provide informed consent. They then watched an instructional video explaining the task, the two available actions, and their respective consequences. After viewing the video, participants had to correctly identify the full form of RARV. Those who answered incorrectly were not permitted to continue. Participants who answered correctly proceeded to the main part of the survey, which included three embedded attention check questions to maintain data quality.

The study was approved by the Institutional Review Board at the University of Michigan (ID HUM00249731). The workers completed the surveys using Qualtrics. Each of the 12 surveys was completed by at least 10 MTurk workers. In total, we obtained 4092 query responses from 124 workers.

4.2 Data Cleaning

We noticed that the data collected from MTurk workers contained many sub-optimal action choices, likely due to participants either misunderstanding the task or not paying sufficient attention, even though all workers passed the basic attention check questions. To improve data quality, we opted to exclude data from workers who made specific "obviously" sub-optimal choices. These involved two scenarios: (1) Using the RARV when the threat level was 0%, and (2) Not using the RARV when the threat level was 100%, when health was 10, and time was above 10. In the first scenario, choosing not to use the RARV is optimal, as it ensures no loss of health or time, while using the RARV results in unnecessary time loss. In the second scenario, failure to use the RARV is sub-optimal, as the human's health would drop to 0, prematurely ending the mission.

We removed the data from the corresponding MTurk workers, resulting in the removal of data from $83 \cup 10 = 85$ workers. Thus, our final dataset consisted of responses from 39 workers, corresponding to 1287 query responses.

4.3 Data Analysis

A logistic regression was trained using threat level as the predictor and the action choice as the target for each state (h, c) in the query set of the cleaned data. A sample logistic curve is shown in Fig. 2. The point at which the logistic curve reaches a value of 0.5 corresponds to d^* from Eq. 13. Thus, we get the raw reward weights for losing health $w(H, C)$ as a function of health remaining and time remaining at each of the queried states. This is shown as a heatmap in Fig. 3a.

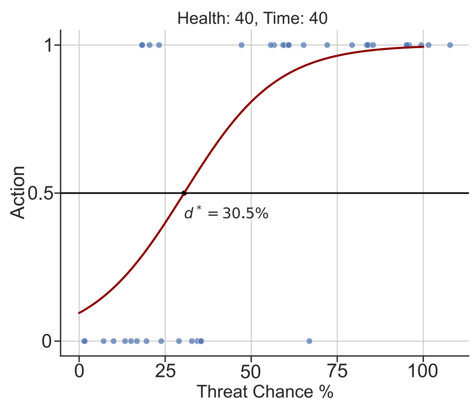


Fig. 2. Sample logistic curve learned from the collected data

In order to get a smooth function for the health reward weight, we used the forward selection method of model selection using the Akaike Information Criterion (AIC). To do this, we incrementally added features from the set $\{H, C, H^2, C^2, HC\}$ to logistic regression models, computed the AIC, and stopped when the AIC increased. This gave us a logistic regression model with $\{H, C\}$ as the features. The final resulting model is given by,

$$w(H, C) = \frac{1}{1 + \exp(0.26H - 0.17C - 0.79)}. \tag{14}$$

A heatmap generated using this model is shown in Fig. 3b. Here, the reward weight increases with increasing value of time remaining and decreasing value of health remaining.

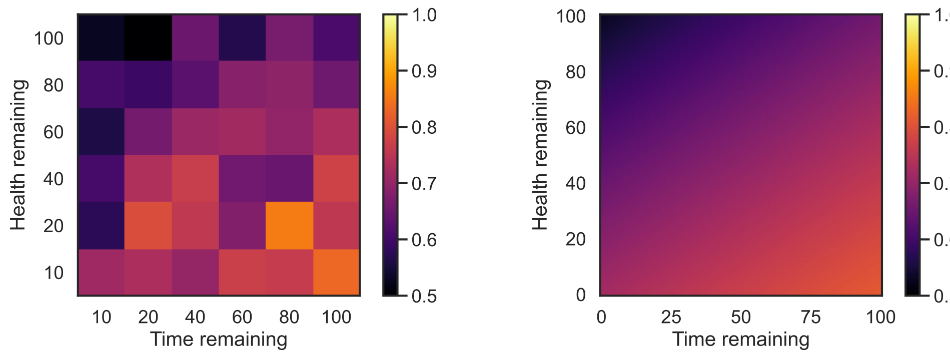
5 Study 2 - Studying the effect of State Dependence of Reward Weights

This section describes the second study where we try to test the effectiveness of a robot using the state dependent reward function compared to one that uses a constant reward function.

In our previous work [9], we found that the constant reward function performed almost as well as a personalized reward function. However, we note that the constant strategy could be too conservative, especially in situations where a higher risk appetite is necessary. For example, in the context of our reconnaissance mission, if very few time points are remaining, the participants must take more risks in order to save time and finish the mission. In such situations, the constant strategy will still suggest conservative actions, disregarding the low time points remaining. However, a robot using our state dependent reward function will suggest riskier actions, helping the participants complete the mission and garnering more trust.

5.1 Testbed and Scenario

We modify the testbed used in our previous work [8] to incorporate the state dependent reward function. The testbed is developed in Unreal Engine 4 and simulates the Intelligence, Surveillance,



(a) Raw data

(b) Smoothed model

Fig. 3. Heatmaps showing (a) raw data of learned health reward weights at each queried state and (b) the smoothed function for the state dependence of reward weights

and Reconnaissance mission scenario. Each mission entails sequentially searching 10 sites for potential threats. The human starts with either 100 or 40 points of health, operationalized as high or low armor and either 100 or 40 points of time, operationalized as time available to complete the mission. The mission is considered to have ended prematurely if the human reaches 0 health or 0 time points at any point during the mission.

Before making their action choice, a drone enters the search site and relays information about the chance of threat present inside the site to the human-robot team. Then, the testbed asks the participant what their initial decision is regarding the choice of action. To reiterate, there are two actions available to the human - USE or NOT USE the Robotic Armored Rescue Vehicle (RARV). Using the RARV results in 10 point loss of time. Encountering a threat without protection from the RARV results in 10 point loss of health. Table 1 shows this in a tabular format.

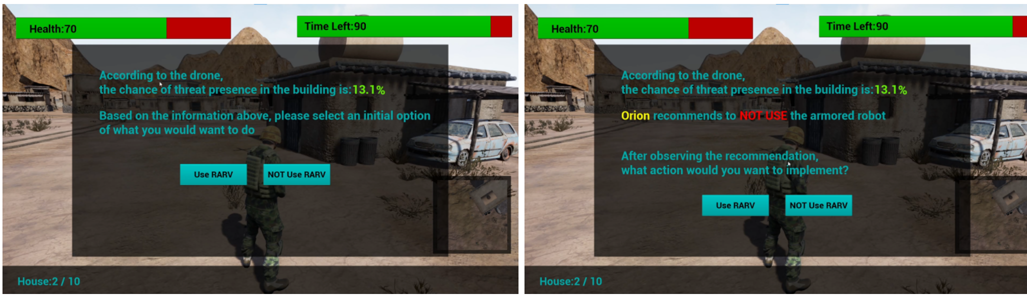
Table 1. Health and Time Loss

Used RARV?	Threat Present?	Health Loss	Time Loss
YES	YES	0	10
YES	NO	0	10
NO	YES	10	0
NO	NO	0	0

After making their initial decision, the participants are shown the recommendation from the intelligent agent. They are then asked for their final decision about which action they want to implement. These interfaces are shown in Fig. 4. The participants are then showed the consequences of their action choice (shown in Fig. **??**) and the appropriate loss of health or time is applied.

The participants are given the objective to complete the mission while maximizing the amount of health points and the amount of time points remaining at the end of the mission. They are also told that the mission will end prematurely if either the human's health or the time remaining drops to 0 at any point during the mission and that their performance bonus will be affected negatively if this happens. This encourages the participants to perform as well as they can.

The testbed also captures the participant's trust and perceived performance of the recommendation after each search site. The interfaces are shown in Fig. 5. It is emphasized that they should

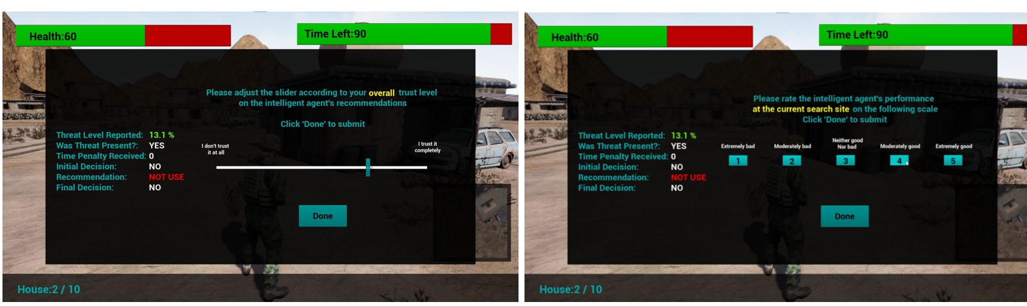


(a) Initial Action Choice

(b) Final Action Choice

Fig. 4. Interfaces to let the participants choose their (a) Initial Action Choice before receiving a recommendation from the intelligent agent and (b) Final Action Choice after receiving the recommendation.

consider their entire interaction history with the intelligent agent when rating their trust and should only consider the current interaction for rating their performance feedback.



(a) Trust feedback slider

(b) Performance feedback scale

Fig. 5. Interfaces to let the participants rate their trust and performance feedback on the intelligent agent's recommendations. The trust feedback slider converts to a value between 0 to 100 with a stepsize of 2.

5.2 Robot Interaction Strategies

We design two interaction strategies for the intelligent agent - a strategy using the state dependent reward weights (Eq. 14) and the other using a constant reward weight of $w = 0.8062$, which is based on our earlier work [9]. The participants were told that they will interact with two different robots - one called Orion and the other called Titan. Behind the scenes, these corresponded to the constant and the state dependent strategies respectively.

5.3 Active Threat Selection

In order to emphasize the different risk appetites of the two robot strategies, we adopt the following strategy for setting the threats and threat levels:

- 50% of the time the threats are set randomly with a probability of 0.6, and the threat level shown to the participant is close to 0.9 if there is a threat present inside the site and close to 0.1 if it is not present inside the site. These values are sampled to show some variance in the reported threat levels to the participants.

- With the remaining 50% chance, the threat levels are set actively to induce a difference between the two interaction strategies. This is done by leveraging the differing d^* values (Eq. 13) for the two strategies. Essentially, we compute d^* for both strategies and sample a value between them as the threat level shown to the participants. Then the presence of threat is sampled from a Bernoulli distribution based on this sampled threat level.

This strategy of setting the threats and threat levels serves two purposes. Firstly, since we set the threats and threat levels completely randomly 50% of the time, the participants are kept in the dark about the underlying manipulation of the threats and threat levels in the remaining 50% cases. Secondly, by setting the threats and threat levels actively 50% of the time, we try to make it so that the constant strategy will recommend risk-averse actions and the state-dependent strategy will recommend riskier actions. This allows us to efficiently show the differences in the risk appetites of the two strategies without having to resort to large amounts of human-subjects data, which will be very difficult to collect. We take inspiration from the active learning literature [11] where the goal is to actively select queries by robots in order to efficiently gain insights into the underlying reward functions of the robot.

5.4 Experiment Design

We designed a 2 (robot strategy) x 2 (vulnerability) mixed design for the human-subjects study. The robot strategy was a within subjects factor while the vulnerability was a between subjects factor. Each participant completed 2 missions, one with each robot strategy. Each mission entailed searching sequentially through 10 search sites. The order for the manipulations was counterbalanced to avoid any learning effects. The details of the manipulations are presented below. The study was approved by the University of Michigan Institutional Review Board under ID HUM00262685.

5.4.1 Robot Strategy. The two interaction strategies presented in Sec. 5.2. These were operationalized for the participants as two different intelligent agents called Orion (constant) and Titan (state dependent). The participants were not told any technical details about the differences between the intelligent agents.

5.4.2 Vulnerability. We had two levels of vulnerability. In the low vulnerability condition, the participants started from a health value of 100 and a time left value of 100. In the high vulnerability condition, the participants started with a health value of 40 and a time left value of 40. These were operationalized as high or low armor and high or low mission time. The participants were notified that the amount of armor only affects the total damage the human can take. Each time the human takes damage, the health loss is 10 points irrespective of armor level.

5.5 Data Collection

We recruited 40 students (Age: 22.18 ± 3.05, 21 Female) from the University of Michigan. The participants came in, signed an informed consent form, and went through a detailed instructional video that briefed them about the experimental procedure. After that, they completed a pre-experimental questionnaire consisting of statistically significant items from Chung and Yang [18]. Then, they completed two missions, one with each intelligent agent. After each mission, they also completed a post-mission questionnaire about their trust in the intelligent agent, reliance intentions, and workload. Finally, the participants were compensated for their time and given any performance bonus. The base compensation was $15 with a performance bonus of up to $10 based on the amount of health and time lost during the mission. This was done to encourage the participants to lose as little health and time as possible.

We removed the data from any participants that ended the mission prematurely (reached either 0 health or time points at any point during the mission), leaving us with a dataset from 33 participants. Of these, 19 belonged to the low vulnerability condition and 14 belonged to the high vulnerability condition.

5.6 Measures

We divide our measures into trust-related and performance-related measures as detailed below.

5.6.1 Trust-related.

- **Average Trust Feedback during interaction:** Measured as the average of the trust feedback given by the participants after each search site in the testbed.
- **End of Mission Trust Feedback:** Measured as the trust feedback given by the participants after searching the last search site in the mission.
- **Post Mission Trust:** Measured using Muir's trust questionnaire Muir and Moray [48], adapted to the context of autonomous systems. This 8-item slider type survey was administered after each mission the participant completed.
- **Reliance Intentions:** Measured using the questionnaire in Lyons and Guznov [42]. This 6-item, 7-point Likert type survey was administered after each mission the participant completed.
- **Average Initial Agreement:** Measured using the average number of times the participants initial decision (before observing the recommendation) matched with the recommendation given by the intelligent agent. This serves as an indicator of how well the recommendation strategy aligns with what the participant would have done in that context without the recommendations.
- **Average Final Agreement:** Measured using the average number of times the participant selected the recommended action as their final choice of action. This serves as an indicator of behavioral trust.

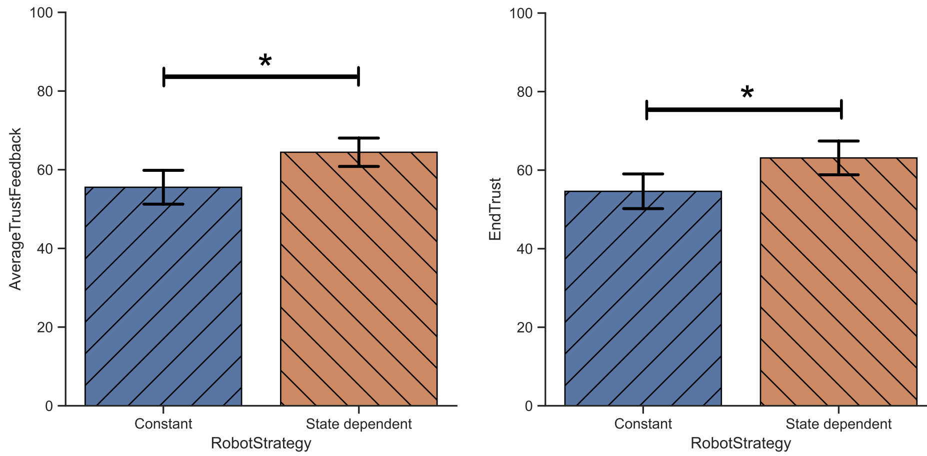
5.6.2 Performance-related.

- **Average Performance Rating:** Measured as the average of the performance feedback given by the participants to the intelligent agent's recommendations during the mission. This serves as an indicator of how well the participants thought the intelligent agent's recommendations were.
- **Percent Points Lost:** Measured as the percentage of total points lost (health + time) with respect to the starting amount of points. A lower amount of points lost indicates better performance.

5.7 Results

This section presents major results from out statistical analyses. Mixed-methods ANOVA was performed to test for statistically significant differences between the interaction strategies and their interactions with the level of vulnerability. The level of significance was set at 0.05. In all the figures in this section, $*$ indicates $p < 0.05$ and $**$ indicates $p < 0.01$.

5.7.1 Average Trust During Interaction. We find a significant main effect of robot strategy ($F(1, 31) = 6.270,\ p = 0.018$) (see Fig. 6a). The state dependent strategy gets rated significantly higher in trust reported during the interaction.



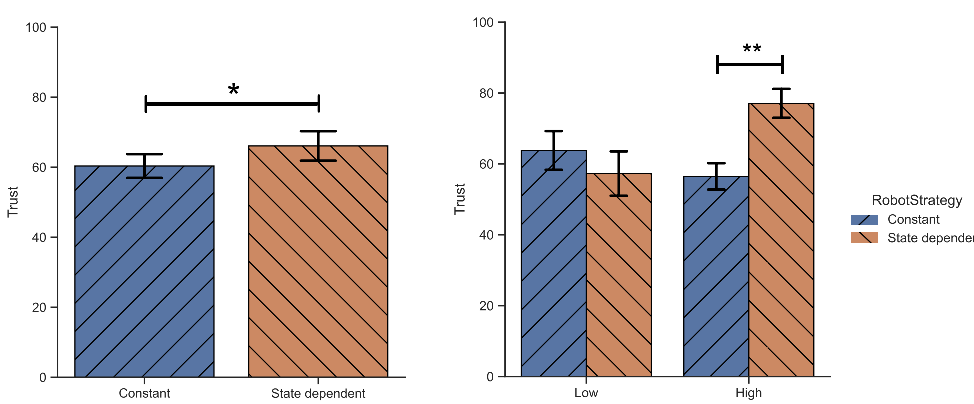
(a) Average Trust

(b) Trust after last site

Fig. 6. Trust measures captured in the testbed: (a) Trust feedback given by the participants averaged over the 10 search sites, (b) Trust feedback given by the participants after searching the last search site.

5.7.2 End-of-mission Trust. We find a significant main effect of robot strategy ($F(1,31) = 5.103,\ p = 0.032$) (see Fig. 6b). The state dependent strategy leads to a higher level of trust towards the end of the mission, showing that it is able to gain the trust of the participants.

5.7.3 Post-mission Trust Survey. We find a significant main effect of robot strategy ($F(1,31) = 5.572,\ p = 0.025$) (see Fig. 7a). The participants report a higher level of trust in the state dependent strategy compared to the constant strategy, showing the alignment with the trust feedback collected in the testbed.



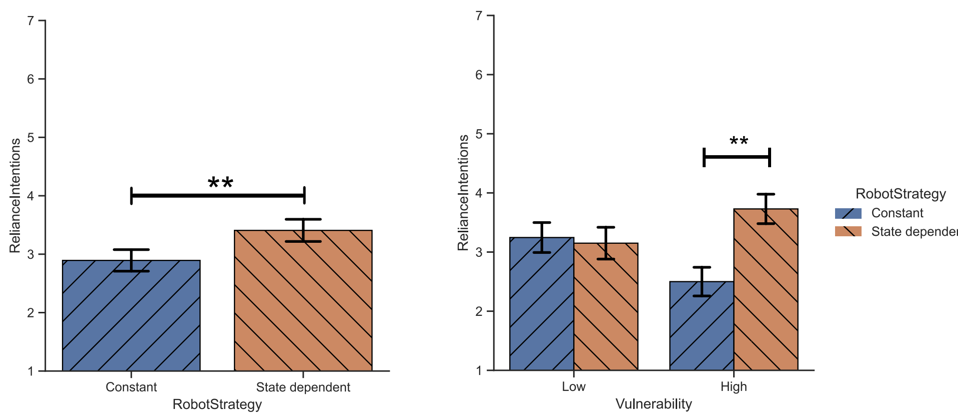
(a) Main Effect

(b) Interaction Effect

Fig. 7. Trust measured using a subjective trust questionnaire [48] after completing a mission. (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.

We also see a significant interaction effect of robot strategy and vulnerability on the reported post-mission trust ($F(1, 31) = 13.210,\ p < 0.001$) (see Fig. 7b). This shows that the gain in trust of the state dependent strategy over the constant strategy is more pronounced in the high vulnerability condition.

5.7.4 Post mission Reliance Intentions. We find a significant main effect of robot strategy on the participants' reported future reliance intentions [42] on the intelligent agent ($F(1, 31) = 9.813,\ p = 0.004$) (see Fig. 8a). The participants would rely on the state dependent strategy more than on the constant strategy in potential future interactions.



(a) Main Effect

(b) Interaction Effect

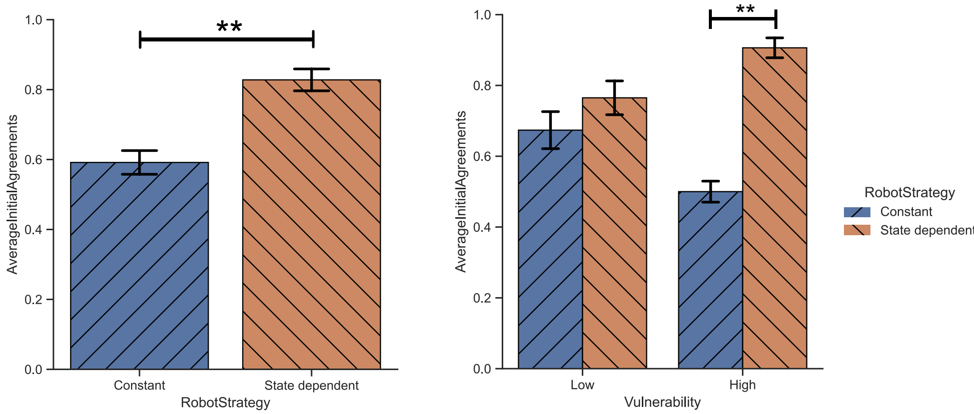
Fig. 8. Reliance intentions measured using a 6-item 7-point Likert scale mentioned in Lyons and Guznov [42]. (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.

We also find a significant interaction effect of robot strategy and vulnerability on future reliance intentions ($F(1, 31) = 10.358,\ p = 0.003$) (see Fig. 8b). We see that the participants report to rely on the state dependent strategy a lot more compared to the constant strategy in the high vulnerability condition.

5.7.5 Average Initial Agreement. We find a significant main effect of robot strategy ($F(1, 31) = 25.530,\ p < 0.001$) on the average agreement between the initial action choice made by the participant and the recommendation given by the intelligent agent (see Fig. 9a). We see that the state dependent strategy results in a lot more agreements than the constant strategy, showing that the reward weights are aligned with the participant's preferences.

We also see a significant interaction effect of robot strategy and vulnerability on the initial agreements ($F(1, 31) = 7.144,\ p = 0.012$). We see that the difference in initial agreements between the constant and the state dependent strategies is much higher in the high vulnerability condition (see Fig. 9b).

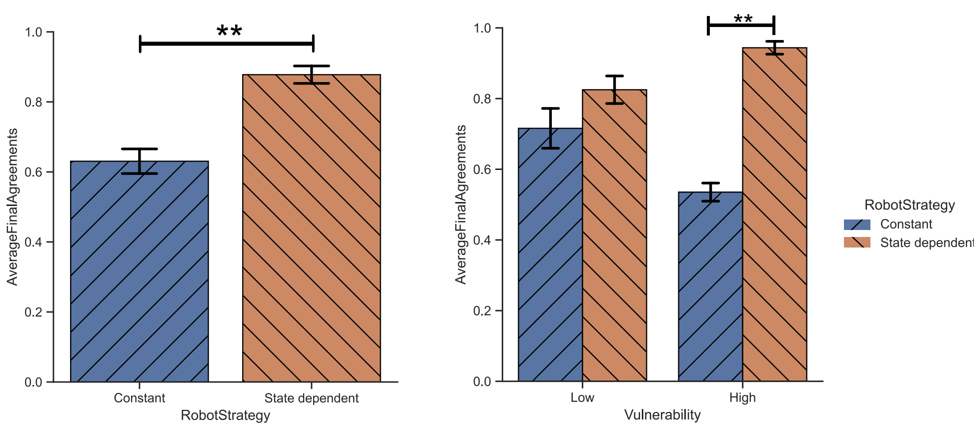
5.7.6 Average Final Agreement. We find a significant main effect of robot strategy ($F(1, 31) = 32.278,\ p < 0.001$) on the average agreement between the final action choice of the participants and the recommendation given by the intelligent agent (see Fig. 10a). Again, the state dependent strategy results in more agreement with what the participants finally decide to do, indicating a higher level of behavioral trust in this strategy compared to the constant strategy.



(a) Main Effect

(b) Interaction Effect

Fig. 9. Agreement between the participants' initial choice of action with the robot's recommendation, averaged over the interaction period. (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.



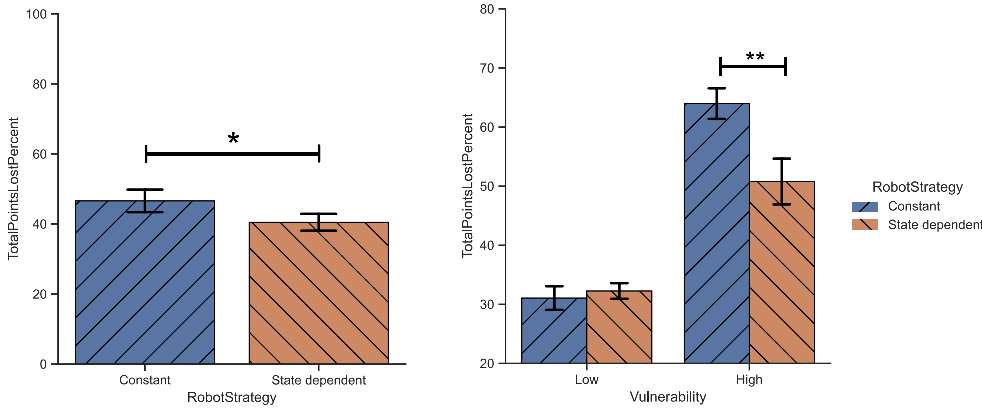
(a) Main Effect

(b) Interaction Effect

Fig. 10. Agreement between the participants' final choice of action with the robot's recommendation, averaged over the interaction period. (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.

We also find a significant interaction effect of robot strategy and vulnerability on the average final agreements ($F(1, 31) = 7.780,\ p = 0.009$). We see that the difference in agreements between the strategies is highest in the high vulnerability condition (see Fig. 10b).

5.7.7 Objective Performance - Percent Points Lost. We find a significant main effect of robot strategy on the percent points lost ($F(1, 31) = 6.666,\ p = 0.015$). The percentage is computed with respect to the total points the human had at the beginning of the mission. A lower value indicates better performance. We see that the state dependent strategy leads to a lower loss in points (see Fig. 11a), indicating that the team performs better when paired with the state dependent strategy.



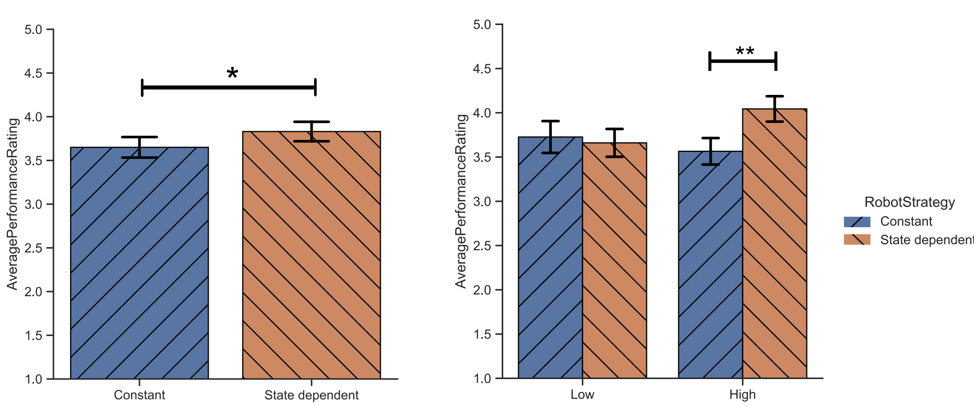
(a) Main Effect

(b) Interaction Effect

Fig. 11. Percent total points lost over the interaction period, taken with respect to the total points the human starts with (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.

There is also a significant interaction effect of robot strategy and vulnerability ($F(1, 31) = 7.993,\ p = 0.008$). We see that under the high vulnerability condition, participants lose a significantly less amount of points when paired with the state dependent strategy (see Fig. 11b).

5.7.8 Subjective Performance - Average Performance Rating. We find a significant main effect of robot strategy ($F(1, 31) = 4.985, p = 0.033$) on the performance rating given to the robot's recommendations by the participants during the interaction, averaged over the mission. As can be seen in Fig. 12a, the participants rate the state dependent strategy's recommendations as better performing than the constant strategy's recommendations.



(a) Main Effect

(b) Interaction Effect

Fig. 12. Subjective performance rating of the intelligent agent captured in the testbed, averaged over the interaction period (a) main effect of robot strategy, (b) interaction effect of robot strategy and vulnerability.

There is also a significant interaction effect of robot strategy and vulnerability ($F(1, 31) = 6.084,\ p = 0.019$) on the average performance rating of recommendations. As seen in Fig. 12b, the difference in performance is more apparent in the high vulnerability condition, with the state dependent strategy being rated higher in performance.

6 Conclusion

In this paper, we studied the effect of learning and using a state-dependent reward function in the context of reconnaissance missions where a robot acts as an action recommender to the human. Towards this, we conducted two studies - one through Amazon Mechanical Turk to collect data and learn the state dependent reward function and another in-person study which tries to see the effects of a robot using this state dependent reward function during interaction on trust and team performance. This study extends our previous work [10] by considering the context of the current state in the reward function within the trust-aware Markov Decision Process.

Our results indicate that the robot using the state dependent strategy results in higher trust, both subjective and behavioral, and better team performance, both objective and subjective, compared to the robot using a constant strategy. This is in line with prior literature like Chiou and Lee [17] where the authors conclude that automation responsivity could be more relevant than reliability and reliance of automated systems. Görür et al. [29] also study the effect of adapting to both short term human behavior and long term human characteristics by collaborative robots. Their results also indicate that such adaptivity increases the efficiency and naturalness of collaboration and human trust. In our scenario, we incorporate responsivity through the state dependent reward function of the robot - the risk appetite of the state dependent strategy changes based on the current state of the human-robot team.

Further, we see that these trust and performance benefits are more pronounced in the situation where the vulnerability is higher, which is in line with out own prior work [10] where we found that aligning the robot's reward function with that of the human is more important when risks are higher. These results are also in line with the recommendations of the meta analysis by Mckenna et al. [46], where the authors claim that the vulnerability to the human in Human-Robot Teaming studies could be an important predictor of trust development in the robot. Finally, our results align with that of Stower et al. [63] where the authors find that risky robot behavior led to higher trust compared to a robot that was consistently risk-averse but suboptimal. Our state-dependent strategy had a generally higher risk appetite compared to the constant strategy, and it led to higher trust by the human teammates.

We conclude this paper by discussing some limitations in this work and possible directions of future research that could try and address these limitations.

6.1 Limitations and Future Work

We have focused on a dyadic human-robot interaction where one human interacts with one robot. In real life, however, teams are usually larger and consist of multiple humans and robots Schneiders et al. [60]. Future work should try and address this limitation by incorporating multi-agent trust models (e.g. [28, 54, 67]) into a decision-making framework. Some very recent work has started to tackle this problem [19, 23, 31, 33, 58] but a lot more work needs to be done toward solving this problem.

We learned a reward function and demonstrated its effectiveness in a context where the conflicting objectives were about losing health and losing mission time. Most people tend to bias themselves toward saving the human's health, thus resulting in the reward weight for losing health being greater than 0.5 at all states (Fig. 14). Given that, we cannot claim that our results will directly translate to another context where there may be a larger variance in the preferences of the humans about the objective. We encourage researchers to study this problem in different contexts.

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